

3/12/25

RESTART

Q) Suppose Host A wants us to send a large file to Host B. The path from Host A to Host B has three routes of rates $R_1 = 500 \text{ kbps}$, $R_2 = 2 \text{ Mbps}$, & $R_3 = 1 \text{ Mbps}$.

(i) Assume no other traffic in the network, what is the throughput for the file transfer?

(ii) Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?

(iii) Repeat (i) and (ii), but with R_2 reduced to 100 kbps.

[3 marks]

A: (i)

Throughput is limited by the slowest link on the path, and transfer time is file size divided by its throughput.

$$R_1 = 500 \text{ kbps} ; R_2 = 2 \text{ Mbps} = 2000 \text{ kbps} ; R_3 = 1 \text{ Mbps} = 1000 \text{ kbps}$$

$\therefore$ End-to-End throughput in a path of links is the minimum of the link rates.

$$\therefore \text{throughput} = \min(500, 1000, 2000)$$

$$= \underline{\underline{500 \text{ kbps}}}$$

(ii)

time to send 4 million bytes with 500 kbps

$$\text{file size} : 4 \times 10^6 \text{ bytes} = 4 \times 8 \times 10^6 \quad (1 \text{ byte} = 8 \text{ bits})$$

$$= \underline{\underline{32 \times 10^6 \text{ bits}}}$$

$$\text{throughput} = 500 \text{ kbps} = \underline{\underline{500 \times 10^3 \text{ bits/sec}}}$$

$$\text{transfer time (T)} = \frac{\text{file size}}{\text{throughput}} = \frac{32 \times 10^6}{500 \times 10^3}$$

$$= 0.0640 \times 10^3$$

$$= \underline{\underline{64 \text{ sec}}}$$

(iii)

$$R_1 = 500 \text{ kbps} ; R_2 = 100 \text{ kbps} ; R_3 = 1000 \text{ kbps}$$

$$(i) \text{ throughput} = \underline{\underline{100 \text{ kbps}}} = 100 \times 10^3 \text{ bits/sec}$$

$$(ii) \text{ transfer time (T)} = \frac{32 \times 10^6}{100 \times 10^3} = 0.3200 \times 10^3 = \underline{\underline{320 \text{ sec}}}$$

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Q) Distinguish between packet switching & circuit switching using a clear example. [3 marks]

A:

Aspect	Circuit Switching	Packet Switching
Connection Setup	Requires dedicated path : before data transfer.	No dedicated path : direct data transfer.
Resource Allocation	Resources (bandwidth) are reserved for entire duration	Resources shared among users; allocated on demand.
Data transmission	Entire bandwidth continuously used by single connection.	Data divided into packets; bandwidth shared.
Path	Fixed physical path between source and destination.	No physical paths; packets can take different routes.
Phases	3 Phases : Connection establishment, data transfer, connection release.	Only data transfer phase.
Bandwidth Utilization	Wastage occurs when circuit is idle.	Efficient; bandwidth is shared.
Transmission Method	Not store-and-forward.	Store-and-forward technique.
Delay	Uniform delay between data units.	Non-uniform delay (varies based on network congestion).
Reliability	More Reliable	Less Reliable
<u>Congestion</u> → traffic jam	Occurs during connection establishment phase.	Occurs during data transfer phase.
Best for	Real-time Communication Eg:- Voicecalls, traditional telephony	Data Communication Eg:- Email, file transfer, web-browsing

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A:

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Q) Examine why the voice & video traffic is often sent over TCP rather than UDP in today's Internet. [3 marks]

A:

feature	TCP	UDP
firewall traversal	Easy (port 80/443)	often blocked
Reliability	guaranteed delivery	Best-effort only
Congestion Control	Built-in	None
Security (TLS)	supported	Not directly
Transport layer Security		
streaming protocol	HTTP/DASH	RTP

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Q) Analyze the operation of alternating-bit protocol for the cases of loss packets & loss ACK (Acknowledgement) with a neat diagram.

Q

Suppose that the five measured Sample RTT values are 106 ms, 120 ms, 140 ms, 90 ms and 115 ms. Compute the Estimated RTT after each of these Sample RTT values is obtained, using a value of $\alpha = 0.125$ and assuming that the value of Estimated RTT was 100 ms just before first of these five samples were obtained. Compute also the DevRTT after each sample is obtained,

22
13
20
21

76

→ IVth Sem

~~Ith theory~~

$$5 \times 3 = 15$$

$$= +3$$

$$18$$

$$+ 2$$

$$= 20$$

Ch1: Introduction

Q1) Define the Internet as a network of networks

The Internet is a global interconnection of autonomous networks using a common protocol suite to communicate. Each network runs its own administration but agrees to interconnect via standard protocols to enable end-to-end data delivery.

Q2) Describe the role of edge devices versus core routers.

Edge devices :- ① Eg :- hosts, servers, IoT

② generate and consume application data at the network's periphery.

Core routers :- ① forward packets at high speed through the backbone, interconnecting networks to provide wide-area reachability.

Q3) Explain packet switching in contrast to circuit switching.

Packets that share links statistically, improving utilization and scalability. Unlike circuit switching, no dedicated path is reserved; packets may queue at routers, introducing variable delay and possible loss.

Q4) List the primary Internet service types (access & content).

Access services connect end systems to network

Eg :- home broadband, cellular campus networks etc.

Content services include the web, email, streaming, cloud, storage, and interactive platforms delivered over IP.

Q5) Define a protocol and its importance.

A protocol is a set of rules governing message formats and ordering, and actions on transmission and receipt.

Shared protocols enable interoperable communication among independently designed systems.

- Q) Give examples of Internet protocol pairs at different layers.
At the transport layer, TCP provides reliable, connection-oriented service, while UDP offers connectionless, best-effort delivery.
At the network layer, IP provides addressing and routing.
At the link layer, Ethernet and Wi-Fi handle local delivery.
- Q) Describe how TCP provides reliable data transfer.
TCP uses sequence numbers, acknowledgments and retransmission timers to ensure ordered, loss-free delivery. Flow control and congestion control adjust sending rates to protect receivers and the network.
- Q) Explain what best-effort service means for IP.
① Best-effort means the network does not guarantee delivery, order, or timing.
② It attempts to forward packets as resources allow.
③ Reliability and timing assurances, if needed are handled by higher layers or applications.
- Q) Compare client-server and peer-to-peer application architectures.
• Client-server: Centralizes resources in always-on servers with many requesting services.
• Peer-to-peer: Distributes roles among peers that act as both clients and servers, improving scalability but complicating coordination.
- Q) Define throughput and delay as performance metrics.
• Throughput: The rate of successful data delivery over a path, often measured in bits per second.
• Delay: The total time a packet takes end-to-end including processing, queuing, transmission, and propagation components.

Q) List the 4 delay components along a network path.

- ① Processing delay occurs at routers/hosts, queuing delay arises from contention.
- ② Transmission delay depends on packet size and link rate.
- ③ Propagation delay depends on physical distance and medium.
- ④ Total nodal delay is their sum at each hop.

6/12/25

Q) Explain addressing and DNS at a high level.

- Ans:
- End systems use IP addresses to identify interfaces, enabling routing across networks.
 - DNS resolves human-readable names to IP addresses, allowing applications to locate services without memorizing numeric addresses.

Q) Describe the concept of a network edge access network.

An access network connects end users to their first router via technologies like DSL, cable, fiber, Ethernet, or cellular. Its capacity and sharing nature influence user-perceived throughput and delay.

Q) Define sockets as the application's interface to transport.

A socket is the API endpoint through which applications send and receive data via the transport layer. It binds to local ports and, with addresses, identifies communication endpoints for TCP or UDP.

Q) Summarize layered architecture benefits in the Internet.

Layering decomposes functionality, enabling modular design, independent evolution, and interoperability. Each layer offers services to the layer above while hiding its internal implementation behind well-defined interfaces.

Q) Explain the difference between hosts and end systems.

Hosts, also called end systems, are the devices where applications run (PCs, servers, smartphones). They sit at the edge of the Internet and communicate over the network core via access networks.

Q1) Describe the role of the network core.

The network consists of interconnected routers and high-speed links responsible for forwarding packets. Its primary function is to move data between end systems by making per-packet routing and forwarding decisions.

Q2) Define a packet & its key components.

A packet is a formatted block of data containing a header and a payload. The header carries control information (eg:- source/destination addresses, protocol fields), while the payload carries application data.

Q3) Explain store-and-forward transmission in routers.

Routers receive an entire packet, store it briefly in a buffer, and then forward it to the next hop. This introduces per-hop transmission, propagation and queuing delays that accumulate along the path.

Q4) List common types of access networks.

Typical access networks include residential broadband (DSL, cable, fiber), cellular (4G, 5G), and enterprise WiFi/Ethernet. Each differs in bandwidth, shared medium behavior, and last-mile infrastructure.

Q5) Describe how a link's transmission rate affects delay.

Transmission delay equals packet length divided by link rate. Higher link rates reduce the time to push bits onto the wire, lowering total end-to-end delay.

Q6) Define queuing delay & its causes.

Queuing delay occurs when packets contend for the same output link and must wait in a buffer. It increases with higher traffic intensity and burstiness and can lead to packet loss when buffers overflow.

Q) Explain loss in packet-switched networks.

Loss happens when a router's buffer is full and arriving packets are dropped. End systems or protocols may detect loss and retransmit, but this adds delay and variability.

Q) Differentiate between services and protocols in the Internet.

Services specify what the network offers to applications (eg:- reliable data transfer, best-effort delivery). Protocols define how entities communicate to provide those services, including message formats and rules for exchange.

Q) Define a network protocol & its components.

A protocol is an agreement on message syntax, semantics, and timing between communicating entities. It governs how state is established, messages are interpreted, and actions are triggered on events.

Q) Describe the client-server application architecture.

In client-server, a persistent server with a fixed address provides services while client initiate contact. Clients are typically transient and do not directly communicate with each other for the service.

Q) Explain the socket interface from an application's perspective.

A socket is the API through which an application sends and receives messages into the network. It abstracts lower-layer details while exposing necessary parameters like destination address and transport protocol.

Ch. Networking & Routing

Q) Define the role of the network layer in the Internet stack.

The network layer provides end-to-end packet delivery across multiple links by performing routing and logical addressing. It encapsulates segments into packets, applies forwarding decisions at routers, and remains agnostic to transport-layer semantics.

Q) Explain IPv4 addressing structure and notation.

— An IPv4 address is a 32-bit value written in dotted-decimal.

Eg:- 192.168.1.10

— It consists of a network prefix and a host identifier; the split is indicated by a subnet mask or CIDR prefix length.

Eg:- /24

Q) Describe subnetting and why it is used.

Subnetting divides a larger network into smaller subnets by extending the network prefix eg:- 10.0.0.0/8 into /16 or /24 blocks. It improves address utilization, limits broadcast domains and enables hierarchical routing within organizations.

Q) Contrast supernetting (CIDR aggregation) with subnetting.

Supernetting aggregates contiguous prefixes into a shorter prefix to reduce routing

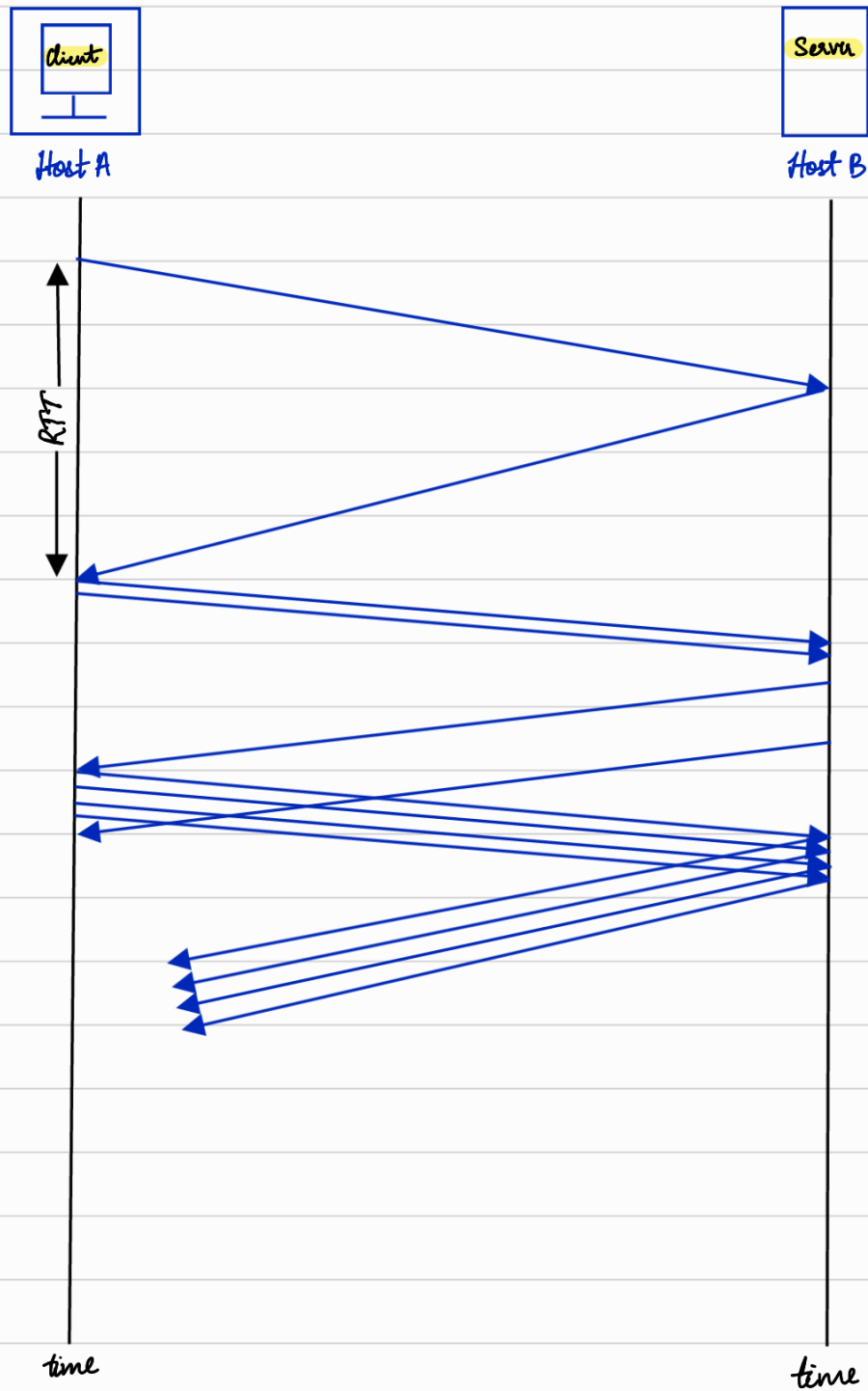
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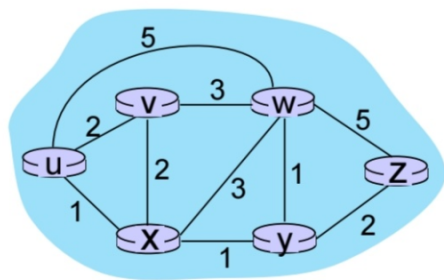
- reading the entire 'cn-merged-ensemble'.poff file.
- read gfg on based Q's.

Q) Illustrate TCP slow start component of TCP congestion-control algorithm with neat diagram.

A:



Q)



Solve using Dijkstra starting from u.

$$D(b) = \min. [D(b), D(a) + C_{a,b}]$$

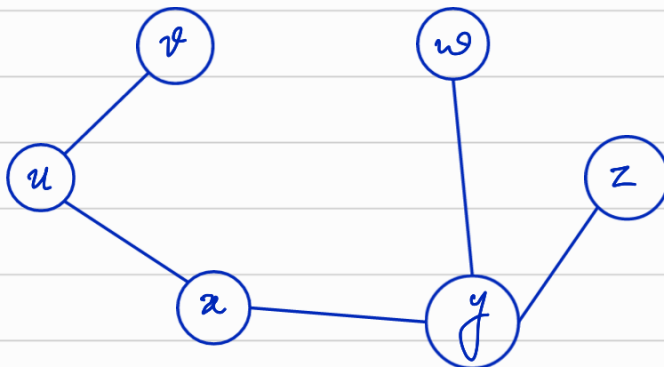
$$5, 1+3$$

$$\infty, 1+1$$

A:

computing table :-

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	0, u	5, w	1, x	∞	∞
1	ux	0, u	4, x		2, x	∞
2	uxy	0, u	3, y			4, y
3	uxyz		3, y			4, y
4	uxyvw					4, y
5	uxyvwz					



$\Rightarrow$ least-cost-path tree from u

resulting forwarding table in u $\Rightarrow$

dest.	outgoing link
v	u,v $\rightarrow$ direct
x	u,x
y	u,x
w	u,x
z	u,x

other via x

